

SOLVING STRESS URINARY INCONTINENCE (SUI) THROUGH ADAPTIVE IMPLANT DESIGN

Team SoFIT (Solutions for Urinary Incontinence)

WHY IT MATTERS

Every leak steals **dignity and quality of life**. For millions of women, their **social and physical worlds shrink** because of urinary incontinence.

Approximately **45%** of adult women in the United States experience urinary incontinence.

NEED STATEMENT

Women with SUI need an **adaptive way to address the underlying mechanism of leakage during daily activities** so that they can live independently without worrying about their continence.

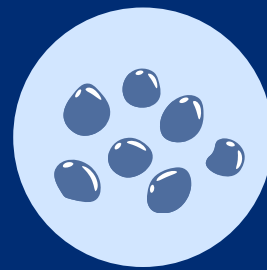


ISSUE WITH EXISTING SURGICAL SOLUTIONS: LACK OF DYNAMICS



No Pressure Control

Current solutions exert constant pressure around the urethra, which can cause tissue erosion and increase the risk of urinary tract infections (UTIs).



No Anatomical Adaptability

Current solutions are not able to adapt to changes in tissue caused by tissue erosion, making them less effective over time.



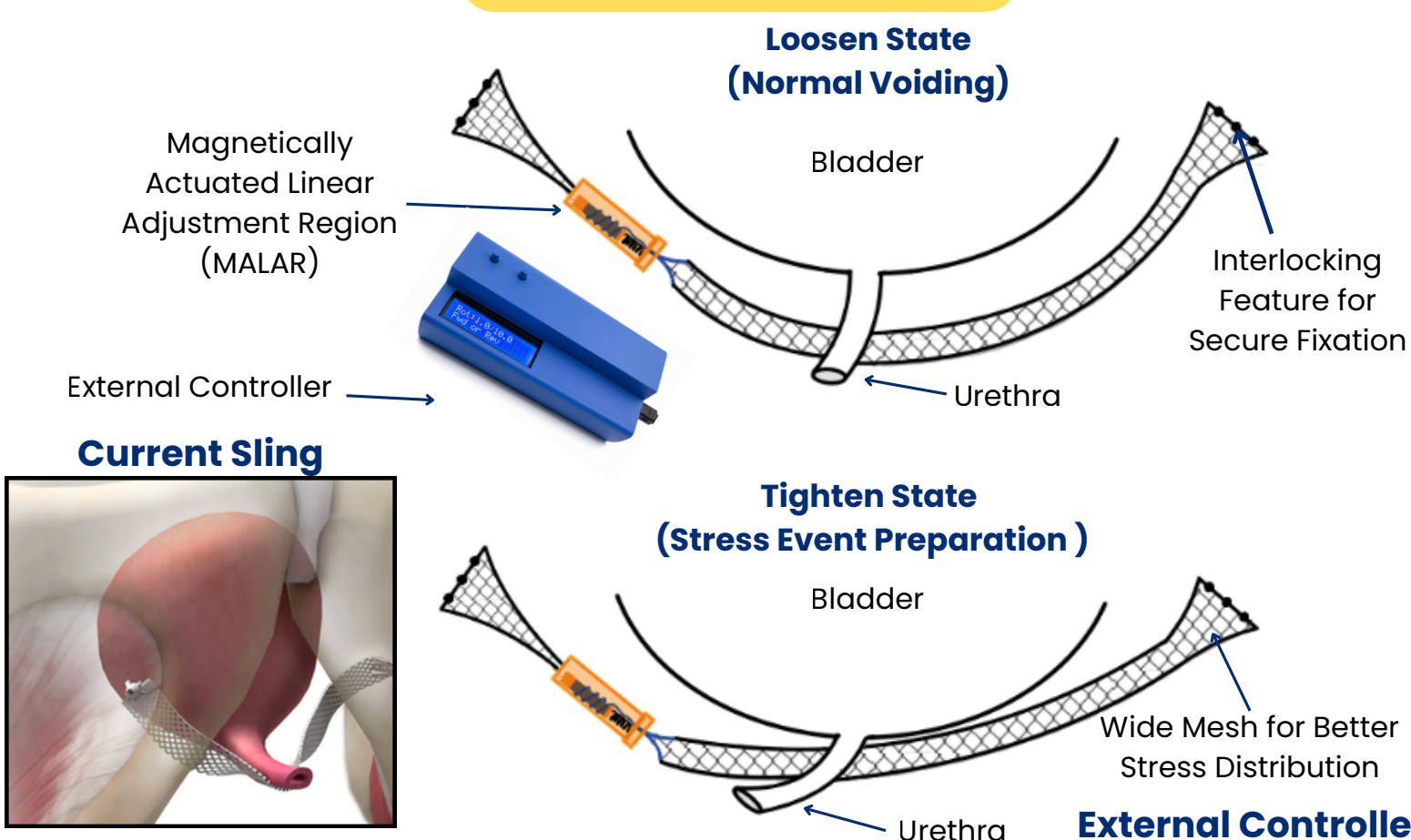
No Responsiveness to Physiological Change

Current solutions are not able to adapt to changes in patient weight, making them less effective if changes occur.

OUR SOLUTIONS

MagFit Dynamic: A Magnetically Adjustable Urethral Sling

Overview of the Solution

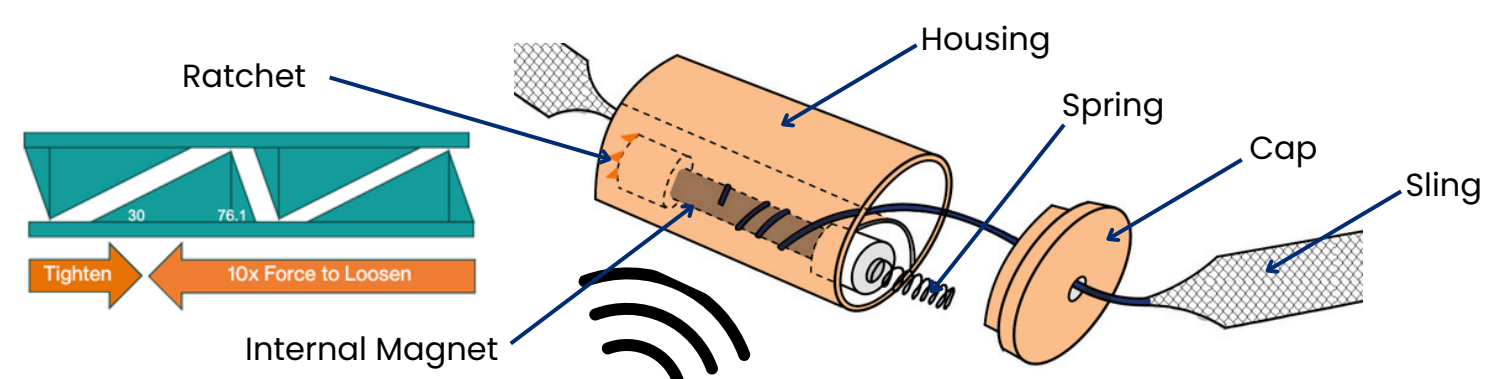


MALAR Actuation

1. External magnet rotates
2. Internal magnet rotates (magnetic coupling)
3. Attached mesh (string) winds/unwinds
4. Linear displacement generated

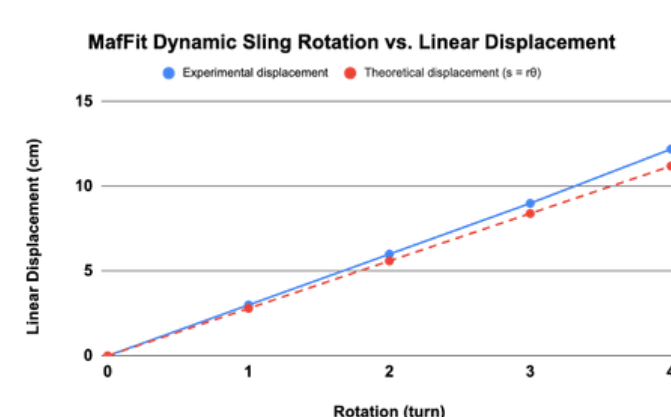


Magnetically Actuated Linear Adjustment Region (MALAR)



Testing Results

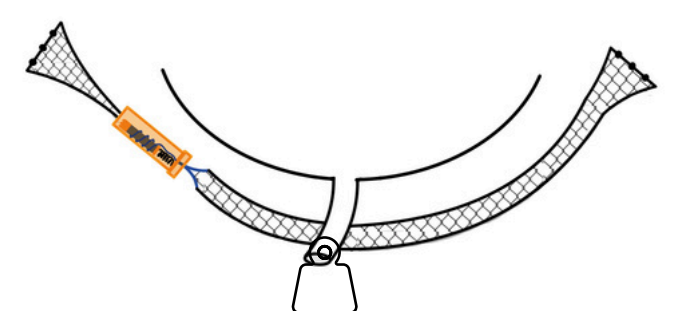
1. Actuation Characterization



Rotation-displacement relationship ($D=9\text{mm}$)

- ~3.0 cm/turn
- Error: ~5.7 % (compared to theoretical values)
- Scaled down to implantable magnets ($D=4\text{mm}$)
- Expected: 1.19 cm/turn

2. Load Bearing Capacity



MagFit Sling maximum load capacity

- 60 g
- equates to ~156 cm H₂O in urethra
- Compared to target pressure load
- Target: 60-70 cm H₂O